Dopamine and deep brain stimulation accelerate the neural dynamics of volitional action in Parkinson's disease

Paper: Koehler et al. (2024). Published human electrophysiology study.

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Page references below use PDF page numbers, including front matter.

To which general question does this paper relate?

Measurement, impaired movement initiation, and the control objective. Relates treatment effects to the interval between preparatory neural activity and muscle activation.

How does it answer the general questions?

Measured. Sensorimotor ECoG, STN-LFP and EMG were recorded during self-initiated movements in 25 patients. Readiness potentials and classifiers estimated preparatory activity; directional spectral analyses assessed cortico-subthalamic communication (PDF pp. 1–8).

Treatment effect. Dopaminergic medication and STN-DBS shortened preparation-to-execution latencies and shifted directional coupling towards theta. These findings associate treatment benefit with changes in the timing and frequency structure of communication; they do not establish beta-power minimization as a sufficient objective (PDF pp. 8–11).

Model relevance. Movement-initiation latency is a useful behavioural outcome. These latencies should not be substituted directly for axonal delays in the notebook model: they summarize distributed preparation and execution processes. No nonlinear delayed-state model or feedback controller is identified.

Version. This is the published counterpart of note 05; the two files are not independent studies.

What papers does it cite that could provide more information?

- Kornhuber and Deecke (1965), work on the *Bereitschaftspotential*: origin of the preparatory-potential measure (ref. 2, PDF p. 11).
- Lofredi et al. (2018), *Dopamine-dependent scaling of subthalamic gamma bursts with movement velocity in patients with Parkinson's disease*: another movement-related outcome (ref. 56, PDF p. 12).